A3-PS003-TC429

Federated Traffic-Flow Prediction with Communication-Cost Evaluation

(Urban Flow)

AIORI Hackathon 3 | Project Report

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October 14, 2026

Abstract

This project implements a simplified federated-learning experiment for short-horizon traffic-flow prediction using the METR-LA dataset. The target is the next traffic reading, estimated from the preceding five readings. A baseline model is compared with a federated setup that simulates three clients, performs local stochastic-gradient training, and aggregates model parameters over ten rounds. The reported test results are close: baseline MAE 1.406789, RMSE 2.556219 and R^2 0.511770; federated MAE 1.406638, RMSE 2.553150 and R^2 0.512941. The experiment therefore shows a very small improvement in the tested run, not a large or general advantage. The estimated model-parameter communication is 1,440 bytes uploaded and 1,440 bytes downloaded, or 2,880 bytes (2.812 KiB) total, excluding protocol overhead. The report documents the problem, background, related work, proposed method, optimization, architecture, data flow, timeline and limitations.

Problem Description

- Traffic conditions change over time, so a model must use recent readings to estimate the next reading.
- Centralizing raw traffic observations can create data-sharing and governance concerns in multi-owner deployments.
- Federated learning can train a shared model from local client updates without requiring raw client training records to be centrally pooled.
- Model quality must be assessed using consistent test data and quantitative metrics.
- Communication overhead matters because clients exchange model parameters during each federated round.

Background

Traffic-flow forecasting uses historical sensor readings to estimate future conditions. In this implementation, one traffic series is selected from the cleaned METR-LA data and transformed into supervised examples: the previous five values form the input and the next value is the target. The data is split chronologically, preserving the time order and reducing the risk of training on future readings when evaluating past-to-future prediction. Features are standardized using statistics fitted on the training set.

Federated learning is a distributed training paradigm in which participating clients train locally and share model updates or parameters for aggregation. This project simulates that workflow in one Python process: the training set is split among three clients, each client trains an SGDRegressor initialized from the current global parameters, and the coordinator calculates a sample-count-weighted average of the resulting coefficients and intercepts. This is a simplified simulation; it is not a deployment across separate devices or a secure federated-learning framework.

Scope and assumptions

The current implementation is a controlled experiment on a selected traffic series. Three clients are simulated by splitting the training examples; there is no actual network transport, encryption, differential privacy, or secure aggregation. The communication numbers assume six model parameters (five coefficients plus one intercept) represented as 8-byte values, with one upload and one download per client per round.

Literature & RFC Review

No specific RFC is required by the current implementation; this review focuses on the most relevant federated-learning research and the dataset/model tooling.

McMahan et al. (2017) — Communication-Efficient Learning of Deep Networks from Decentralized Data.

Introduced Federated Averaging (FedAvg), a widely used approach that aggregates locally computed model updates. This project follows the high-level idea of local training followed by weighted aggregation, but uses a linear SGDRegressor and a simplified in-process simulation.

Kairouz et al. (2021) — Advances and Open Problems in Federated Learning.

Surveys the practical challenges of federated learning, including communication, statistical heterogeneity, privacy, systems constraints and evaluation. These issues motivate reporting both predictive metrics and a communication estimate.

Li et al. (2020) — Federated Optimization in Heterogeneous Networks.

Discusses optimization challenges when client data and systems differ. The current experiment uses a simple split of one time-ordered training set; it does not fully model the diversity or intermittent connectivity of real traffic agencies.

METR-LA dataset (commonly distributed with DCRNN research).

A traffic-sensor dataset used in traffic forecasting research. This project uses the supplied METR-LA CSV and a selected sensor series; the results depend on that preprocessing and sensor choice.

Proposed Solution

The system follows a centralized coordinator with simulated clients. Clients train locally and share model parameters for weighted aggregation. All clients run in the same Python process in this experiment.

1. Load the cleaned traffic dataset and select the configured sensor series.
2. Create rolling windows of five consecutive readings and use the next reading as the label.
3. Split examples in chronological order into training and test sets, then standardize the input features using training data only.
4. Train/evaluate a baseline model and simulate three federated clients on partitions of the training data.
5. For each of ten rounds, initialize each client model from the current global coefficients/intercept, train locally for five epochs, then average client parameters weighted by local sample counts.
6. Generate predictions on the held-out test set, calculate MAE/RMSE/R^2, and export predictions, metric comparisons and communication-cost tables.

Outputs

The scripts export prediction CSVs, model_comparison.csv, communication_cost.csv, client_communication_comparison.csv and PNG visualizations.

Solution Optimization

Reported results show a very small change in metrics; they do not establish a substantial improvement.

Aspect	Baseline / before	Federated experiment / after
Training arrangement	Single training model	Training partitioned across 3 simulated clients
Training coordination	No client aggregation	10 rounds of weighted parameter aggregation
Prediction MAE	1.406789	1.406638
Prediction RMSE	2.556219	2.553150
R^2 score	0.511770	0.512941
Communication	No federated exchange in baseline	Estimated 1,440 B upload + 1,440 B download

Communication-cost calculation

Parameters per model = 5 coefficients + 1 intercept = 6. Estimated upload = 3 × 6 × 8 × 10 = 1,440 bytes; download is estimated identically. Total = 2,880 bytes (2.8125 KiB before display rounding), excluding protocol overhead.

Architecture Design (UML-style System Diagram)

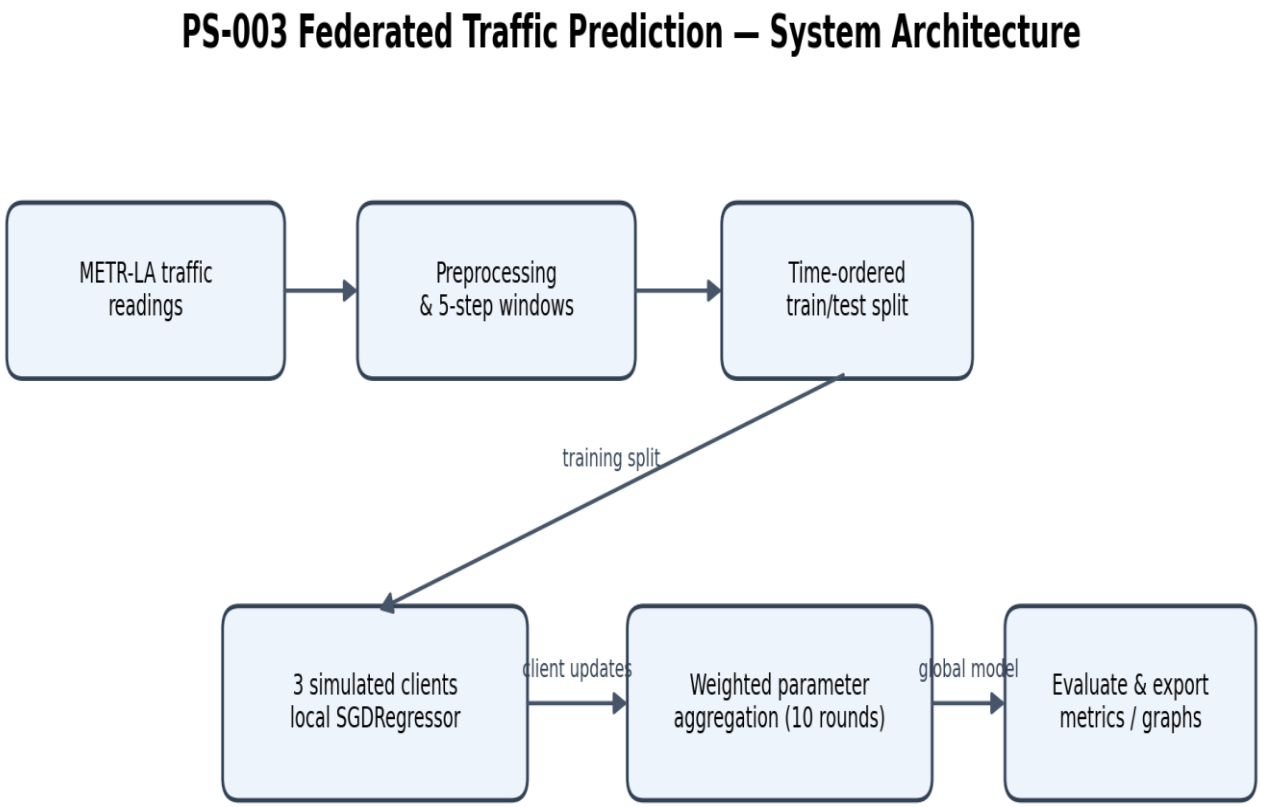


Figure 1. System architecture of the simplified federated traffic-prediction experiment.

Data Flow Diagram

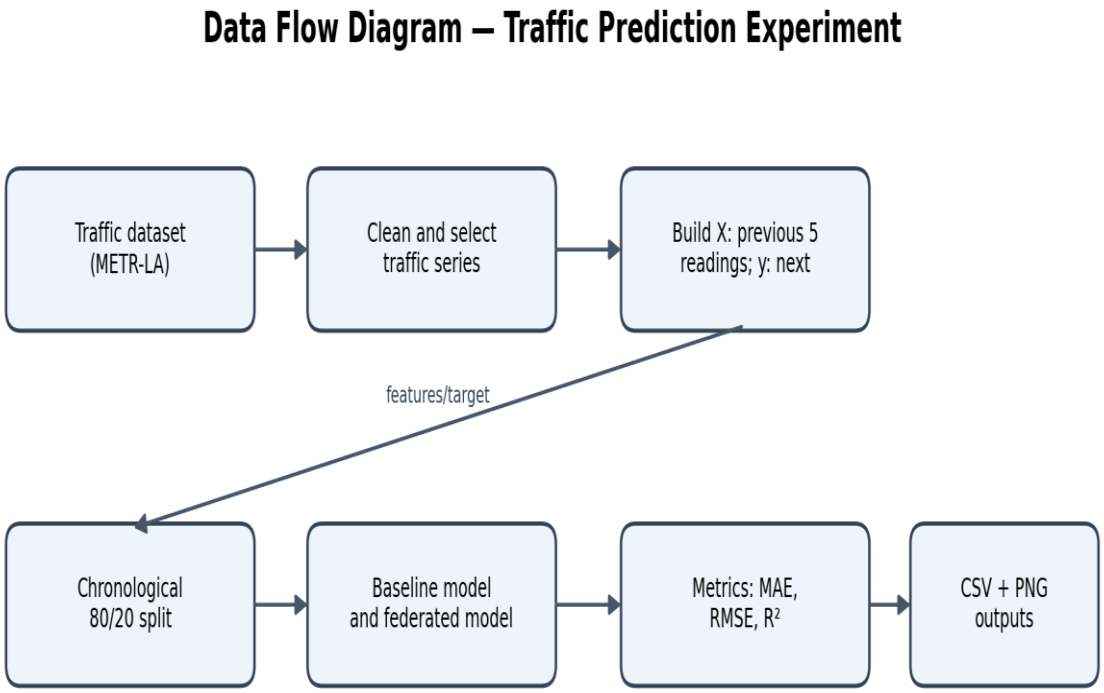


Figure 2. Data flow from METR-LA input to prediction metrics and output files.

Deployment Timeline

The table reflects stages and status indicated by the current project files and reported command outputs.

Milestone	Activity	Status
1. Data preparation	Load/clean traffic series and form 5-reading windows	Completed in current project
2. Baseline	Train baseline and save predictions	Output files present; rerun to reproduce if needed
3. Federated training	3 clients, 10 rounds, 5 local epochs	Completed in reported run
4. Evaluation	Compare MAE, RMSE and R^2 on test set	Results exported
5. Communication analysis	Estimate parameter transfer and client-count comparison	Estimate and graph generated
6. Packaging	README, requirements, report, code, CSVs and graphs	Prepared; verify final ZIP before submission

Reproducibility checklist

- Install dependencies from requirements.txt.
- Run client_comparison.py and communication_graph.py.
- Run federated_learning.py to regenerate outputs.
- Run final_comparison_graph.py and final_results_report.py.
- Confirm output CSVs and PNGs correspond to the latest run.
- Replace cover placeholders with actual team/member details.

Bibliography

[1] H. B. McMahan et al., “Communication-Efficient Learning of Deep Networks from Decentralized Data,” AISTATS, 2017.

[2] P. Kairouz et al., “Advances and Open Problems in Federated Learning,” Foundations and Trends in Machine Learning, 2021.

[3] T. Li et al., “Federated Optimization in Heterogeneous Networks,” Proceedings of MLSys, 2020.

[4] Y. Li et al., “Diffusion Convolutional Recurrent Neural Network: Data-Driven Traffic Forecasting,” ICLR, 2018 (METR-LA dataset context).

[5] Scikit-learn documentation, SGDRegressor and StandardScaler, https://scikit-learn.org/stable/

Limitations and future work

The experiment uses a simple linear model, one selected sensor series and simulated clients in one process. Future work can test multiple sensors, non-IID partitions, stronger baselines, multiple random seeds, actual serialized/network bytes and privacy-preserving mechanisms.

Thank You